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RESEARCH FOCUS

I study the **interplay between representation learning and generative modeling**:

1. How does the **encoding distribution** $q(z_0 | x)$ shape the **learnability of generative models**?
2. How does the **forward process** $p_F(z_{0:T})$ shape the **learnability of generative models**?
3. How can **representation-aware objectives** enable **few-step or one-step generation**?

These questions guide my current focus on **end-to-end (single-step) latent generative models**: jointly learning representations and generators to achieve both **efficient learning and few-step generation**.

EDUCATION AND RESEARCH VISIT

2025 - present	Ph.D. Student <i>School of Engineering, Westlake University</i> Advisor: Prof. Tailin Wu
2023 - 2025	Research Visit <i>Australian National University</i> Worked with Prof. Yuan-Sen Ting
2018 - 2023	Undergraduate, Astronomy <i>Nanjing University</i>

RESEARCH HIGHLIGHT

Feature Information Dynamics in Diffusion

NeurIPS 2026

Diffusion models are often said to generate coarse structure before fine detail. We asked whether this intuition still holds when data are mapped into different representation spaces.

- Using the I-MMSE relation, we defined **feature information density** to measure when class, mask, and Canny information becomes accessible during denoising.
- We found that **representation reshapes the order of feature emergence** across pixel, SDVAE, VAAE, and RAE spaces. Only RAE follows the class, then mask, then Canny order.
- Diffusion models also converge fastest in RAE among these spaces. We hypothesize that its **ordered feature dynamics** helps explain this difference.

SELECTED PUBLICATIONS

Feature Information Dynamics in Diffusion

NeurIPS 2026

Jia-Shu Pan, Tao Zhang, Yufei Huang, Yanjun Sheng, Tailin Wu.

VFScale: Intrinsic Reasoning through Verifier-Free Test-time Scalable Diffusion Model

ICLR 2026

Tao Zhang*, **Jia-Shu Pan***, Ruiqi Feng, Tailin Wu.

* Equal contribution.

Uses the diffusion model's intrinsic energy as a verifier, together with hybrid Monte Carlo Tree Search, to scale reasoning at test time.

REFEREES

Prof. Tailin Wu Westlake University	Assistant Professor wutailin@westlake.edu.cn
Prof. Yuan-Sen Ting Ohio State University	Associate Professor ting.74@osu.edu

CV design adapted from Prof. Yuan-Sen Ting's curriculum vitae.